JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2026 6th April Evening Shift

Question Paper with Answer Key

75	ENGLISH	LIGHT
QUESTIONS	LANGUAGE	FORMAT

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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Which of the following contain the same number of atoms?

(Given : Molar mass in gmol^{-1} of $\mathrm{H}, \mathrm{He}, \mathrm{O}$ and S are 1, 4, 16 and 32 respectively)

A. 2 g of O_2 gas

B. 4 g of SO_2 gas

C. 1400 mL of O_2 at STP

D. $0.05\ \mathrm{L}$ of He at STP

E. $0.0625\ \mathrm{mol}$ of H_2 gas

Choose the correct answer from the options given below :

(A) A and B only **(B)** B and C only **(C)** C and D only **(D)** A, C and E only

Q2. MCQ (Single Correct Answer)

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The Bohr radius of a hydrogen like species is 70.53 pm . The species and the stationary state (n) are respectively

(Given : Hydrogen atom Bohr radius is 52.9 pm)

(A) $\mathrm{Li}^{2+}, 3$ **(B)** $\mathrm{He}^{+}, 3$ **(C)** $\mathrm{He}^{+}, 2$ **(D)** $\mathrm{Li}^{2+}, 2$

Q3. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : The number of compounds among $\mathrm{SO}_2, \mathrm{SO}_3, \mathrm{SF}_4, \mathrm{SF}_6$ and $\mathrm{H}_2\mathrm{S}$ in which sulphur does not obey the Octet rule is 3 .

Statement II : Among $[\mathrm{H}_2\mathrm{O}, \mathrm{ClF}_3, \mathrm{SF}_4]$, $[\mathrm{NH}_3, \mathrm{BrF}_5, \mathrm{SF}_4]$, $[\mathrm{BrF}_5, \mathrm{ClF}_3, \mathrm{XeF}_4]$ and $[\mathrm{XeF}_4, \mathrm{ClF}_3, \mathrm{H}_2\mathrm{O}]$, the number of sets in which all the molecules have one lone pair of electrons on the central atom is 1 .

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q4. MCQ (Single Correct Answer)

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Match List - I with List - II.

Given V_1 and V_2 are initial and final volumes respectively.

List - I (Isothermal process)		List - II (Expression)	
A.	Reversible expansion	I.	$q = 0$
B.	Free expansion	II.	$q = nRT \ln \frac{V_2}{V_1}$
C.	Irreversible Compression	III.	$w = -p_{ext}\,(V_1 - V_2)$
D.	Cyclic reversible	IV.	$\frac{q_{rev}}{T} = 0$

Choose the correct answer from the options given below :

(A) A-II, B-III, C-I, D-IV

(B) A-II, B-I, C-IV, D-III

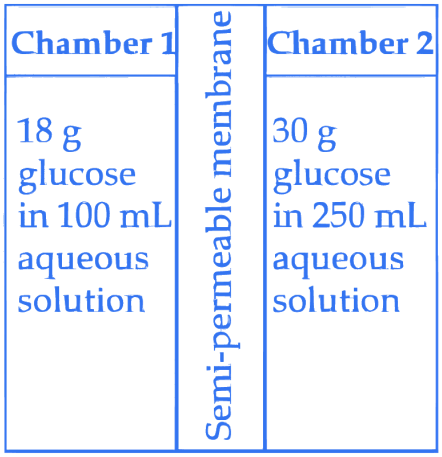
(C) A-II, B-I, C-III, D-IV

(D) A-I, B-II, C-III, D-IV

Q5. MCQ (Single Correct Answer)

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Given below are two statements :



Statement I : H_2O molecules move from the chamber 1 to chamber 2 .

Statement II : The osmotic pressure of a solution prepared by dissolving 50 mg of potassium sulphate (molar mass $= 174\ g/mol$) in 2 L of water (at $27^\circ C$) is 0.0107 bar. (Given: $R = 0.083 dm^3$ bar $K^{-1}\ mol^{-1}$ and assume complete dissociation of electrolyte)

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q6. MCQ (Single Correct Answer)

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Given is a concentrated solution of a weak electrolyte A_xB_y of concentration ' c ' and dissociation constant ' K '. The degree of dissociation is given by :

(A) $\left[\mathrm{K} \times \mathrm{c}^{x+y-1} x^x y^y\right]^{x+y}$

(B) $\left(\frac{\mathrm{K}}{\mathrm{c}^{x+y-1} x^x y^y}\right)^{\frac{1}{x+y}}$

(C) $\left(\frac{\mathrm{c}^{x+y-1} x^x y^y}{\mathrm{K}}\right)^{x+y}$

(D) $\left(\frac{c^{x+y-1} x^x y^y}{\mathrm{K}}\right)^{\frac{1}{x+y}}$

Q7. MCQ (Single Correct Answer)

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For a general redox reaction

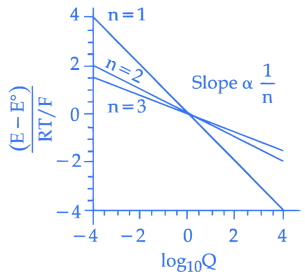
Anode $\quad \mathrm{Red}_1 \rightarrow \mathrm{Ox}_1^{\mathrm{n}_1^+} + \mathrm{n}_1\mathrm{e}^-$

Cathode $\mathrm{Ox}_2 + \mathrm{n}_2\mathrm{e}^- \rightarrow \mathrm{Red}_2^{\mathrm{n}_2^-}$

Which of the following statement is incorrect?

(A) The overall reaction can be written as $\mathrm{n}_2\,\mathrm{Red}_1 + \mathrm{n}_1\mathrm{Ox}_2 \rightleftharpoons \mathrm{n}_2\mathrm{Ox}_1^{\mathrm{n}_1^+} + \mathrm{n}_1\,\mathrm{Red}_2^{\mathrm{n}_2^-}$

(B) The electrons do not appear in the overall reaction because electrons produced at the anode are consumed at the cathode.



(C) Here n is the number of electrons transferred in redox reaction.

(D) If the reaction is carried out reversibly, the electrical work done is equal to the ratio of charge and potential difference through which charge is moved.

Q8. MCQ (Single Correct Answer)

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In a period, the first ionisation enthalpy of the element at extreme left and the negative electron gain enthalpy of the extreme right element, except noble gases, are respectively.

(A) lowest and lowest

(B) highest and lowest

(C) lowest and highest

(D) highest and highest

Q9. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : $F_2O < H_2O < Cl_2O$ is the correct trend in terms of bond angle.

Statement II : SiF_4, SnF_4 and PbF_4 are ionic in nature.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q10. MCQ (Single Correct Answer)

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The correct order of first $(\Delta_i H_1)$ and second $(\Delta_i H_2)$ ionisation enthalpy values of Cr and Mn are :

A. $\Delta_i H_1 : Cr > Mn$

B. $\Delta_i H_2 : Cr > Mn$

C. $\Delta_i H_1 : Mn > Cr$

D. $\Delta_i H_2 : Mn > Cr$

Choose the correct answer from the options given below :

(A) A and B only **(B)** B and C only **(C)** A and D only **(D)** C and D only

Q11. MCQ (Single Correct Answer)

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Which of the following sequences of hybridisation, geometry and magnetic nature are correct for the given coordination compounds?

A. $[NiCl_4]^{2-} - sp^3$, tetrahedral, paramagnetic

B. $[Ni(NH_3)_6]^{2+} - sp^3 d^2$, octahedral, paramagnetic

C. $[Ni(CO)_4] - sp^3$, tetrahedral, paramagnetic

D. $[Ni(CN)_4]^{2-} - dsp^2$, square planar, diamagnetic

Choose the correct answer from the options given below :

(A) A, B, C and D **(B)** B, C and D only **(C)** A, C and D only **(D)** A, B and D only

Q12. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in alcohol, due to differential solubility.

Statement II : Rose essence from rose petals is seperated by steam distillation due to its high volatility and insolubility in H_2O.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

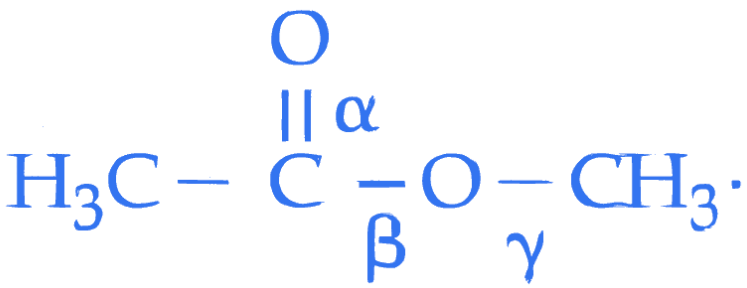
(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q13. MCQ (Single Correct Answer)

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Shown below is the structure of methyl acetate with three different α, β and γ carbon - oxygen bonds.



The correct order of bond lengths of these bonds is :

(A) $\alpha > \beta > \gamma$ (B) $\alpha < \beta < \gamma$ (C) $\alpha = \beta = \gamma$ (D) $\alpha < \beta = \gamma$

Q14. MCQ (Single Correct Answer)

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' x ' is the product which is obtained by the hydrolysis of prop-1-yne in the presence of mercuric sulphate under dilute acidic medium at 333 K . ' y ' is the product which is obtained by the reaction of ethane nitrile with methyl magnesium bromide in dry ether followed by hydrolysis. IUPAC name of product obtained from ' x ' and ' y ' in the presence of barium hydroxide followed by heating is :

(A) 2-Methylpent-4-en-3-one

(B) 4-Methylpent-3-en-2-one

(C) 4-Methylpent-1-ene

(D) 2-Methylpent-3-one

Q15. MCQ (Single Correct Answer)

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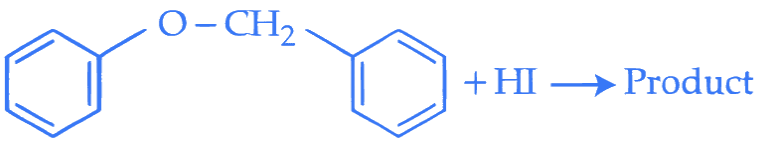
An optically active alkyl bromide C_4H_9Br, reacts with ethanolic KOH to form major compound $[A]$ which reacts with bromine to give compound $[B]$. Compound $[B]$ reacts with ethanolic KOH and sodamide to give compound [C]. One molecule of water adds to compound [C] on warming with mercuric sulphate and dilute sulphuric acid at 333 K to form compound [D]. The functional group in compound D will be confirmed by :

(A) Haloform test
(B) Lucas test
(C) Silver mirror test
(D) Benedict test

Q16. MCQ (Single Correct Answer)

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Consider the following reaction.



Statement I : In the above reaction, product formed will be a mixture of benzyl alcohol and iodobenzene.

Statement II : In the above reaction, the $-O-CH_2-$ bond is cleaved to give the product.

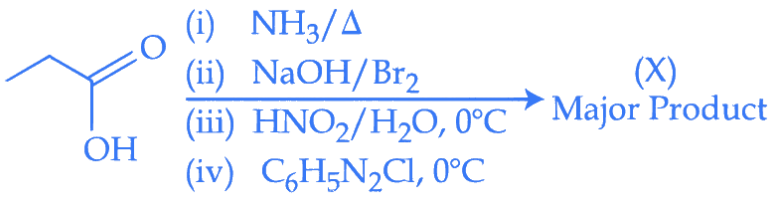
In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

Q17. MCQ (Single Correct Answer)

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Consider the following organic reaction sequence. Choose the final product (X) from the following (consider the major product in all intermediate reactions)

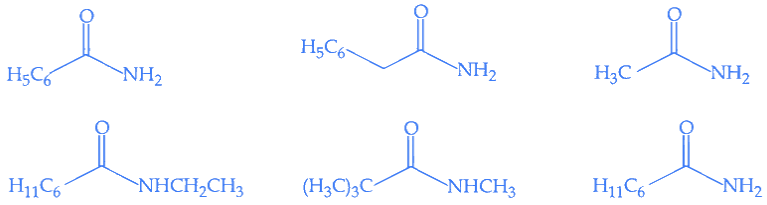


(A) Benzene
(B) Phenol
(C) Propanol
(D) Chlorobenzene

Q18. MCQ (Single Correct Answer)

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The number of compounds from the following which can undergo reaction with Br_2/KOH (alcoholic) to give respective products and these respective products can also be obtained separately by Gabriel phthalimide reaction is :

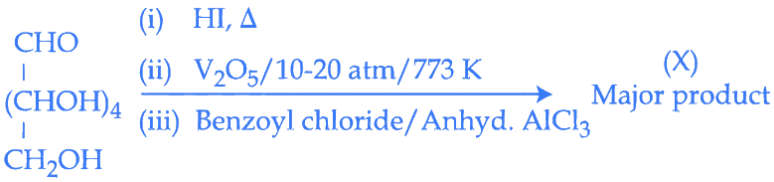


(A) 5 (B) 4 (C) 3 (D) 6

Q19. MCQ (Single Correct Answer)

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Consider the following reactions. Total number of electrons in the π bonds and lone pair of electrons in the product (X) is :



(A) 12 (B) 16 (C) 14 (D) 18

Q20. MCQ (Single Correct Answer)

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Treatment of a gas ' X ' with a freshly prepared ferrous sulphate solution gives a compound ' Y ' as a brown ring. The compounds X and Y are:

(A) NO and $[Fe(NO)]SO_4$

(B) NO_2 and $[Fe\,(NO_2)]SO_4$

(C) N_2O and $[Fe\,(N_2O)]SO_4$

(D) N_2O_4 and $[Fe\,(N_2O_4)]SO_4$

Q21. Numerical

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An excess of $AgNO_3$ is added to 100 mL of a 0.05 M solution of tetraaquadichloridochromium (III) chloride. The number of moles of AgCl precipitated will be ____ $\times 10^{-3}$.

(Nearest integer)

Q22. Numerical

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An alkane (Y) requires 8 moles of oxygen for complete combustion and on chlorination with $Cl_2/h\nu$, (Y) gives only one monochlorinated product (Z). The total number of primary carbon atoms in (Y) is ____ .

Q23. Numerical

500 mL of $0.2\mathrm{M}\mathrm{MnO}_4^-$ solution in basic medium when mixed with 500 mL of 1.5 M KI solution, oxidises iodide ions to liberate molecular iodine. This liberated iodine is then titrated with a standard $x\mathrm{M}$ thiosulphate solution in presence of starch till the end point. If 300 mL of thiosulphate was consumed, then the value of x is ____.

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Q24. Numerical

In a closed flask at 600 K , one mole of $\mathrm{X}_2\mathrm{Y}_4(\mathrm{~g})$ attains equilibrium as given below :

$\mathrm{X}_2\mathrm{Y}_4(\mathrm{~g}) \rightleftharpoons 2\mathrm{XY}_2(\mathrm{~g})$

At equilibrium, $75\%\mathrm{X}_2\mathrm{Y}_4(\mathrm{~g})$ was dissociated and the total pressure is 1 atm . The magnitude of $\Delta_{\mathrm{r}}\mathrm{G}^{\ominus}$ (in kJmol^{-1}) at this temperature is ____ . (Nearest Integer)

(Given : $\mathrm{R}=8.3 \mathrm{~J} \mathrm{~mol}^{-1} \mathrm{~K}^{-1} ; \ln 10=2.3, \log 2=0.3, \log 3=0.48, \log 5=0.69, \log 7=0.84$)

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Q25. Numerical

Decomposition of a hydrocarbon follows the equation $\mathrm{k}=\left(5.5 \times 10^{11} \mathrm{~s}^{-1}\right) \mathrm{e}^{\frac{-28000 \mathrm{~K}}{\mathrm{T}}}$. The activation energy of reaction is ____ kJmol^{-1}. (Nearest Integer)

Given : $\mathrm{R}=8.3 \mathrm{~J} \mathrm{~K}^{-1} \mathrm{~mol}^{-1}$

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined as $f(x)=\frac{2x^2-3x+2}{3x^2+x+3}$. Then f is :

(A) both one-one and onto
(B) one-one but not onto
(C) onto but not one-one
(D) neither one-one nor onto

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Q27. MCQ (Single Correct Answer)

Consider the quadratic equation $\left(n^2-2n+2\right)x^2-3x+\left(n^2-2n+2\right)^2=0, n \in \mathbf{R}$. Let α be the minimum value of the product of its roots and β be the maximum value of the sum of its roots. Then the sum of the first six terms of the G.P., whose first term is α and the common ratio is $\frac{\alpha}{\beta}$, is :

(A) $\frac{61}{37}$
(B) $\frac{121}{81}$
(C) $\frac{364}{243}$
(D) $\frac{1093}{729}$

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Q28. MCQ (Single Correct Answer)

Let $S=\left\{z \in \mathbb{C}: z^2+\sqrt{6} i z-3=0\right\}$. Then $\sum_{z \in S} z^8$ is equal to :

(A) 162
(B) 184
(C) 262
(D) 324

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Q29. MCQ (Single Correct Answer)

The sum of all possible values of $\theta \in [0, 2\pi]$, for which the system of equations :

$x\cos 3\theta - 8y - 12z = 0$

$x\cos 2\theta + 3y + 3z = 0$

$x + y + 3z = 0$

has a non-trivial solution, is equal to :

(A) π (B) 2π (C) 3π (D) 4π

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Q30. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $B = [b_{ij}], 1 \le i, j \le 3$. If $B = A^{99} - I$, then the value of $\frac{b_{31} - b_{21}}{b_{32}}$ is :

(A) 99 (B) 199 (C) 149 (D) 159

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Q31. MCQ (Single Correct Answer)

The sum $1 + \frac{1}{2}(1^2 + 2^2) + \frac{1}{3}(1^2 + 2^2 + 3^2) + \ldots$ upto 10 terms is equal to :

(A) 130 (B) 155 (C) $\frac{315}{2}$ (D) $\frac{325}{2}$

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Q32. MCQ (Single Correct Answer)

A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the 10th floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the lift does not stop at the first and the second floors, is equal to :

(A) 2184 (B) 3064 (C) 7056 (D) 11340

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Q33. MCQ (Single Correct Answer)

Let the mean and the variance of seven observations $2, 4, \alpha, 8, \beta, 12, 14, \alpha < \beta$, be 8 and 16 respectively. Then the quadratic equation whose roots are $3\alpha + 2$ and $2\beta + 1$ is :

(A) $x^2 - 35x + 306 = 0$ (B) $x^2 - 41x + 420 = 0$

(C) $x^2 - 45x + 506 = 0$ (D) $x^2 - 37x + 342 = 0$

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Q34. MCQ (Single Correct Answer)

A bag contains 6 blue and 6 green balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each drawn pair consists of one blue and one green ball is :

(A) $\frac{63}{925}$ (B) $\frac{17}{231}$ (C) $\frac{16}{231}$ (D) $\frac{64}{925}$

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Q35. MCQ (Single Correct Answer)

Let C be a circle having centre in the first quadrant and touching the x-axis at a distance of 3 units from the origin. If the circle C has an intercept of length $6\sqrt{3}$ on y-axis, then the length of the chord of the circle C on the line $x - y = 3$ is :

(A) 8 (B) 6 (C) $6\sqrt{2}$ (D) $8\sqrt{2}$

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Q36. MCQ (Single Correct Answer)

The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm\frac{4\sqrt{6}}{3}$. Let $\mathrm{H} : \frac{x^2}{\mathrm{a}^2} - \frac{y^2}{\mathrm{b}^2} = 1$ be a hyperbola whose eccentricity is equal to the length of semi-major axis of E , and whose length of latus rectum is equal to the length of minor axis of E . Then the distance between the foci of H is :

(A) $\frac{4\sqrt{2}}{\sqrt{7}}$ (B) $\frac{4\sqrt{2}}{7}$ (C) $\frac{4}{\sqrt{7}}$ (D) $\frac{8}{7}$

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Q37. MCQ (Single Correct Answer)

Let $x = 9$ be a directrix of an ellipse E , whose centre is at the origin and eccentricity is $\frac{1}{3}$. Let $\mathrm{P}(\alpha, 0)$, $\alpha > 0$, be a focus of E and AB be a chord passing through P . Then the locus of the mid point of AB is :

(A) $9y^2 = 8x(1 - x)$ (B) $3y^2 = 4x(1 - x)$ (C) $9y^2 = 8x(x - 1)$ (D) $3y^2 = 4x(x - 1)$

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Q38. MCQ (Single Correct Answer)

If $\sin\left(\tan^{-1}(x\sqrt{2})\right) = \cot\left(\sin^{-1}\sqrt{1 - x^2}\right), x \in (0, 1)$, then the value of x is:

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{5}{8}$

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Q39. MCQ (Single Correct Answer)

The shortest distance between the lines $\frac{x-4}{1} = \frac{y-3}{2} = \frac{z-2}{-3}$ and $\frac{x+2}{2} = \frac{y-6}{4} = \frac{z-5}{-5}$ is:

(A) $\frac{5\sqrt{6}}{6}$ (B) $2\sqrt{5}$ (C) $3\sqrt{5}$ (D) $4\sqrt{5}$

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Q40. MCQ (Single Correct Answer)

Let $\vec{\mathrm{a}} = 2\hat{i} + 3\hat{j} + 3\hat{k}$ and $\vec{\mathrm{b}} = 6\hat{i} + 3\hat{j} + 3\hat{k}$. Then the square of the area of the triangle with adjacent sides determined by the vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is :

(A) 450 (B) 900 (C) 1800 (D) 2400

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Q41. MCQ (Single Correct Answer)

Let $\lim\limits_{x \to 2} \frac{(\tan(x-2))(\mathrm{r}x^2 + (\mathrm{p}-2)x - 2\mathrm{p})}{(x-2)^2} = 5$ for some $\mathrm{r}, \mathrm{p} \in \boldsymbol{R}$. If the set of all possible values of q , such that the roots of the equation $\mathrm{r}x^2 - \mathrm{p}x + \mathrm{q} = 0$ lie in $(0, 2)$, be the interval $(\alpha, \beta]$, then $4(\alpha + \beta)$ equals :

(A) 11 (B) 13 (C) 17 (D) 21

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Q42. MCQ (Single Correct Answer)

Let $\mathrm{A}=\begin{bmatrix}1 & 3 & -1\\ 2 & 1 & \alpha\\ 0 & 1 & -1\end{bmatrix}$ be a singular matrix. Let $f(x)=\int_0^x\left(\mathrm{t}^2+2\mathrm{t}+3\right)\mathrm{dt}, x\in[1,\alpha]$. If M and m are respectively the maximum and the minimum values of f in $[1,\alpha]$, then $3(\mathrm{M}-\mathrm{m})$ is equal to :

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(A) 64 (B) 68 (C) 72 (D) 76

Q43. MCQ (Single Correct Answer)

Let $f:\mathbf{R}\to\mathbf{R}$ be such that $f(xy)=f(x)f(y)$, for all $x,y\in\mathbf{R}$ and $f(0)\neq 0$. Let $g:[1,\infty)\to\mathbf{R}$ be a differentiable function such that

$x^2g(x)=\int_1^x\left(\mathrm{t}^2f(\mathrm{t})-\mathrm{tg}(\mathrm{t})\right)dt$

Then $g(2)$ is equal to :

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(A) $\frac{13}{8}$ (B) $\frac{11}{16}$ (C) $\frac{15}{32}$ (D) $\frac{17}{64}$

Q44. MCQ (Single Correct Answer)

The area of the region $\left\{(x,y):x^2-8x\leq y\leq -x\right\}$ is :

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(A) $\frac{343}{6}$ (B) $\frac{637}{6}$ (C) $\frac{437}{6}$ (D) $\frac{523}{6}$

Q45. MCQ (Single Correct Answer)

The value of the integral $\int_{-1}^{1}\left(\frac{x^3+|x|+1}{x^2+2|x|+1}\right)\mathrm{d}x$ is equal to:

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(A) $3\log_e 2$ (B) $2\log_e 2$ (C) $5\log_e 3$ (D) $3\log_e 3$

Q46. Numerical

Let $\mathrm{R}=\{(x,y)\in\mathbf{N}\times\mathbf{N}:\log_e(x+y)\leq 2\}$. Then the minimum number of elements, required to be added in R to make it a transitive relation, is ____ .

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Q47. Numerical

If $\left(1-x^3\right)^{10}=\sum_{\mathrm{r}=0}^{10}\mathrm{a_r}x^{\mathrm{r}}(1-x)^{30-2\mathrm{r}}$, then $\frac{9\mathrm{a_9}}{\mathrm{a_{10}}}$ is equal to ____ .

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Q48. Numerical

Let the line $x-y=4$ intersect the circle $\mathrm{C}:(x-4)^2+(y+3)^2=9$ at the points Q and R . If $\mathrm{P}(\alpha,\beta)$ is a point on C such that $\mathrm{PQ}=\mathrm{PR}$, then $(6\alpha+8\beta)^2$ is equal to ____ .

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Q49. Numerical

Let the image of the point $\mathrm{P}(0,-5,0)$ in the line $\frac{x-1}{2}=\frac{y}{1}=\frac{z+1}{-2}$ be the point R and the image of the point $\mathrm{Q}\left(0,\frac{-1}{2},0\right)$ in the line $\frac{x-1}{-1}=\frac{y+9}{4}=\frac{z+1}{1}$ be the point S . Then the square of the area of the parallelogram PQRS is ____ .

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Q50. Numerical

Let $f(x)=\begin{cases}x^3+8; & x<0,\\ x^2-4; & x\geq 0,\end{cases}$ and $g(x)=\begin{cases}(x-8)^{1/3}; & x<0,\\ (x+4)^{1/2}; & x\geq 0.\end{cases}$

Then the number of points, where the function $g\circ f$ is discontinuous, is ____ .

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

The percentage error in the calculated volume of a sphere, if there is 2% error in its diameter measurement, is ____ .

(A) 1 (B) 2 (C) 6 (D) 8

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Q52. MCQ (Single Correct Answer)

Match List - I with List - II.

List - I		List - II	
A.	Boltzmann constant	I.	$\left[\mathrm{M}^{-1}\mathrm{L}^{3}\mathrm{T}^{-2}\right]$
B.	Stefan's constant	II.	$\left[\mathrm{M}\mathrm{L}^{2}\mathrm{T}^{-1}\right]$
C.	Planck's constant	III.	$\left[\mathrm{M}\mathrm{L}^{2}\mathrm{T}^{-2}\mathrm{K}^{-1}\right]$
D.	Gravitational constant	IV.	$\left[\mathrm{M}\mathrm{L}^{0}\mathrm{T}^{-3}\mathrm{K}^{-4}\right]$

Choose the correct answer from the options given below :

(A) A-I, B-II, C-III, D-IV (B) A-IV, B-III, C-II, D-I

(C) A-III, B-IV, C-II, D-I (D) A-II, B-I, C-IV, D-III

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Q53. MCQ (Single Correct Answer)

A solid sphere (A) of mass $5m$ and a spherical shell (B) of mass m, both having same radius, are placed on a rough surface. When a force of same magnitude is applied tangentially at the highest points of A and B, they start rolling without slipping with an acceleration of a_A and a_B, respectively. The ratio of a_A and a_B is ____ .

(A) $5:21$ (B) $6:10$ (C) $21:25$ (D) $1:5$

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Q54. MCQ (Single Correct Answer)

A body of mass 1 kg moves along a straight line with a velocity $v = 2x^2$. The work done by the body during displacement from $x = 0$ to 5 m is ____ J.

(A) 0 (B) 250 (C) 1250 (D) 1000

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Q55. MCQ (Single Correct Answer)

A cylinder with adiabatic walls is closed at both ends and is divided into two compartments by a frictionless adiabatic piston. Ideal gas is filled in both (left and right) the compartments at same P, V, T. Heating is started from left side until pressure changes to $27\mathrm{P}/8$. If initial volume of each compartment was 9 litres then the final volume in right-hand side compartment is ____ litres. (for this ideal gas $\mathrm{C_P}/\mathrm{C_V} = 1.5$)

(A) 3 (B) 4 (C) 14 (D) 9

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Q56. MCQ (Single Correct Answer)

For an electromagnetic wave propagating through vacuum, $\vec{k}, \vec{E}$ and ω represent propagation vector, electric field and angular frequency, respectively. The magnetic field associated with this wave is represented by:

(A) $\frac{\vec{E}\times\vec{k}}{\omega}$ (B) $\frac{\vec{k}\times\vec{E}}{\omega}$

(C) $\omega(\vec{E}\times\vec{k})$ (D) $\omega(\vec{k}\times\vec{E})$

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Q57. MCQ (Single Correct Answer)

Two identical bodies A and B of equal masses have initial velocities $\overrightarrow{v_1} = 4\hat{i}\,\mathrm{m/s}$ and $\overrightarrow{v_2} = 4\hat{j}\,\mathrm{m/s}$ respectively. The body A has acceleration $\overrightarrow{a_1} = 6\hat{i} + 6\hat{j}\,\mathrm{m/s^2}$ while the acceleration of the other body B is zero. The centre of mass of the two bodies moves in ____ path.

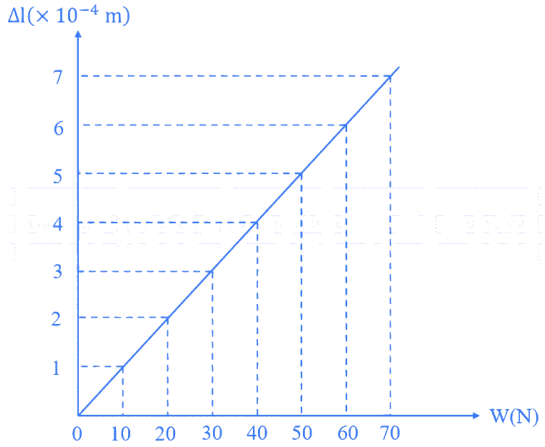
(A) circular (B) parabolic (C) straight line (D) elliptical

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Q58. MCQ (Single Correct Answer)

Figure represents the extension (Δl) of a wire of length 1 meter, suspended from the ceiling of the room at one end with a load W connected to the other end. If the cross-sectional area of the wire is $10^{-5}\ \mathrm{m}^2$ then the Young's modulus of the wire is ____ $\mathrm{N/m^2}$.

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(A) 1.0×10^{11} (B) 2.0×10^{10} (C) 1.0×10^{10} (D) 2.0×10^{11}

Q59. MCQ (Single Correct Answer)

A cylindrical vessel of 40 cm radius is completely filled with water and its capacity is $528\mathrm{dm}^3$ (dm : decimeter) The vessel is placed on a solid block of exactly same height as vessel. If a small hole is made at 70 cm below the top of water level, then horizontal range of water falling on the ground in the beginning is ____ cm .

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(A) $120\sqrt{2}$ (B) $140\sqrt{2}$ (C) $140\sqrt{3}$ (D) $120\sqrt{3}$

Q60. MCQ (Single Correct Answer)

If 2 mole of an ideal monoatomic gas at temperature T, is mixed with 6 mole of another ideal monoatomic gas at temperature $2T$ then the temperature of mixture is:

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(A) $\frac{5}{2}T$ (B) $\frac{5}{4}T$ (C) $\frac{7}{2}T$ (D) $\frac{7}{4}T$

Q61. MCQ (Single Correct Answer)

A spring stretches by 2 mm when it is loaded with a mass of 200 g . From equilibrium position the mass is further pulled down by 2 mm and released. The frequency associated with the system and maximum energy in the spring are ____ Hz and ____ J, respectively.

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(Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

(A) $\frac{5\sqrt{50}}{\pi}$ and 8×10^{-3} (B) $\frac{5\sqrt{50}}{\pi}$ and 8

(C) $10\sqrt{50}$ and 2×10^{-3} (D) $\frac{5\sqrt{50}}{\pi}$ and 16×10^{-3}

Q62. MCQ (Single Correct Answer)

The electric potential as a function of x, y is given by $V = 5\left(x^2 - y^2\right)V$. The electric field at a point $(2, 3)\mathrm{m}$ is ____ $\mathrm{V/m}$.

(A) $(-20\hat{i} + 30\hat{j})$ (B) $(20\hat{i} - 30\hat{j})$ (C) $(20\hat{i} + 45\hat{j})$ (D) $(-4\hat{i} + 6\hat{j})$

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Q63. MCQ (Single Correct Answer)

A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm . The magnetic field at the mid point between the two wires is ____ $\mu\mathrm{T}$.

$\left(\frac{\mu_0}{4\pi} = 10^{-7}\,\mathrm{N/A^2}\right)$

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(A) 30 (B) 300 (C) 150 (D) 0.0

Q64. MCQ (Single Correct Answer)

A square loop of side 2 cm is placed in a time varying magnetic field with magnitude as $B = 0.4\sin(300t)$ Tesla. The normal to the plane of loop makes an angle of 60° with the field. The maximum induced emf produced in the loop is ____ mV .

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(A) 12 (B) 18 (C) 21 (D) 24

Q65. MCQ (Single Correct Answer)

A sphere of capacitance 100 pF is charged to a potential of 100 V . Another identical uncharged metal sphere is brought in contact with the charged sphere, then the change in the total energy stored on these spheres, when they touch is $\alpha \times 10^{-7}$ J. The value of α is ____ .

(combined capacitance of spheres is 200 pF)

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(A) 5 (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

Q66. MCQ (Single Correct Answer)

The energy released if hydrogen atoms are combined to form $^4_2\mathrm{He}$ is ____ MeV .

(Take binding energies per nucleon of $^2_1\mathrm{H}$ and $^4_2\mathrm{He}$ as 1.1 MeV and 7.2 MeV , respectively)

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(A) 6.1 (B) 24.4 (C) 26.6 (D) 5

Q67. MCQ (Single Correct Answer)

Angle of minimum deviation is equal to the half of the angle of prism in an equilateral prism. The refractive index of the prism is ____

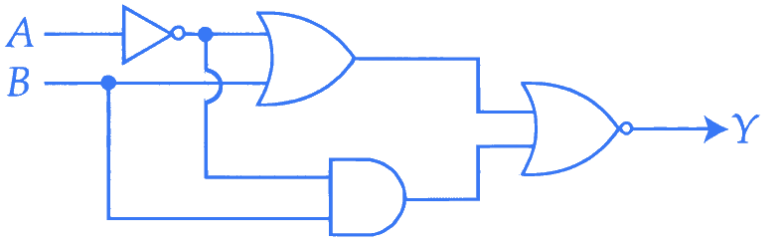
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(A) 1.5 (B) $\sqrt{3}$ (C) $\sqrt{2}$ (D) 1.65

Q68. MCQ (Single Correct Answer)

Refer to the logic circuit given below. For two inputs ($A = 1, B = 1$) and ($A = 0, B = 1$), output (Y) will be ____ .

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(A) 1, 0 respectively

(B) 0, 1 respectively

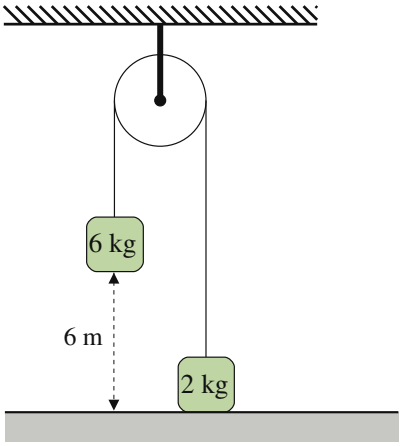
(C) 0, 0 respectively

(D) 1, 1 respectively

Q69. MCQ (Single Correct Answer)

The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6 m above the ground is ____ $\mathrm{m/s}$. Assume pulley is massless and string is light and inextensible. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

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(A) 7.74

(B) 7.20

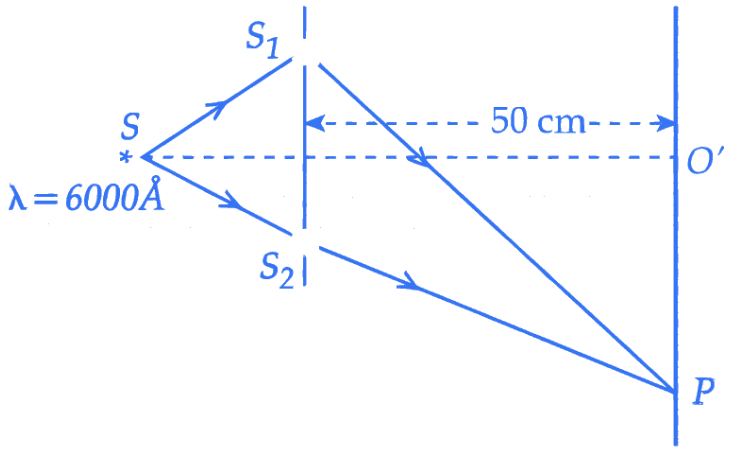
(C) 6.55

(D) 4.50

Q70. MCQ (Single Correct Answer)

In a Young double slit experiment, the wavelength of incident light is $6000\mathring{A}$, the separation between slits S_1 and S_2 is 5 cm and the distance between slits plane and screen is 50 cm , as shown in the figure below. If the resultant intensity at P is equal to the intensity due to individual slits, the path difference between interfering waves is ____ Å.

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(A) 4000 (B) 3000 (C) 2000 (D) 1000

Q71. Numerical

A block takes t time to slide down a plane inclined at 45° to the horizontal. If the surface is made smooth (frictionless), the block takes time $\frac{t}{2}$ to slide down the plane. The coefficient of friction between the block and the inclined plane is $\left(\frac{\alpha}{100}\right)$. The value of α is ____ .

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Q72. Numerical

The de Broglie wavelength for an electron accelerated through the potential difference of V_1 volt is λ_1. When the potential difference is changed to V_2 volt, the associated de Broglie wavelength is increased by 50%. If $(V_1/V_2) = (9/\alpha)$, then the value of α is ____.

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Q73. Numerical

A moving coil of galvanometer when shunted with 2Ω resistance gives a full scale deflection for a current of 500 mA . When a resistance of 470Ω is connected in series it gives a full scale deflection for 10 V potential applied on it. The value of resistance of galvanometer coil is ____ Ω.

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Q74. Numerical

Two cells of emfs 1 V and 2 V and internal resistance 2Ω and 1Ω, respectively connected in parallel, gave a current of 1 A through an external resistance. If the polarity of one cell is reversed, then value of current through the external resistance will be $\frac{\alpha}{5}$ A. The value of α is ____.

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Q75. Numerical

A concave mirror of focal length 10 cm forms an image which is double the size of object when the object is placed at two different positions. The distance between the two positions of the object is ____ cm.

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Answer Key

Chemistry

1. **D**	2. **D**	3. **D**	4. **C**	5. **D**
6. **B**	7. **D**	8. **C**	9. **C**	10. **B**
11. **D**	12. **A**	13. **B**	14. **B**	15. **A**
16. **D**	17. **A**	18. **C**	19. **D**	20. **A**
21. **5**	22. **4**	23. **1**	24. **8**	25. **232**

Mathematics

26. **D**	27. **C**	28. **A**	29. **D**	30. **C**
31. **C**	32. **C**	33. **B**	34. **C**	35. **C**
36. **D**	37. **A**	38. **A**	39. **C**	40. **C**
41. **C**	42. **B**	43. **C**	44. **A**	45. **B**
46. **15**	47. **30**	48. **18**	49. **162**	50. **3**

Physics

51. **C**	52. **C**	53. **A**	54. **C**	55. **B**
56. **B**	57. **C**	58. **C**	59. **B**	60. **D**
61. **A**	62. **A**	63. **B**	64. **D**	65. **B**
66. **B**	67. **C**	68. **C**	69. **A**	70. **C**
71. **75**	72. **4**	73. **50**	74. **3**	75. **10**

For numerical answers given as a range, any value within the range is correct.

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